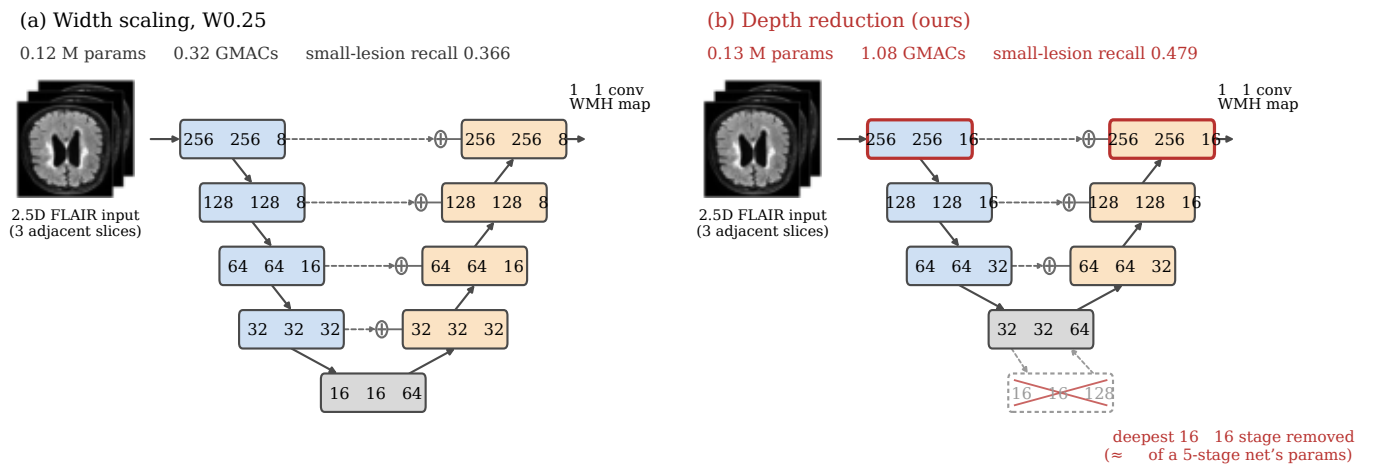
(a) Width scaling, W0.25
0.12 M params
0.32 GMACs
small-lesion recall 0.366
2.5D FLAIR input
(3 adjacent slices)
256 256 8
128 128 8
64 64 16
32 32 32
16 16 64
256 256 8
128 128 8
64 64 16
32 32 32
1 1 conv
WMH map
(b) Depth reduction (ours)
0.13 M params
1.08 GMACs
small-lesion recall 0.479
2.5D FLAIR input
(3 adjacent slices)
256 256 16
128 128 16
64 64 32
32 32 64
16 16 128
256 256 16
128 128 16
64 64 32
1 1 conv
WMH map
deepest 16 16 stage removed
(≈ of a 5-stage net's params)
all five stages kept;
every stage narrowed to 0.25 the baseline
saved budget moved to the high-resolution
stages (2 the channels of W0.25)
encoder block: 2 (conv 3 3 – IN – LeakyReLU)
decoder block: up-conv, concat, 2 conv
bottleneck (deepest stage)
skip connection (channel concatenation)